An Agentic System for Evidence-Grounded Digestive Disease Diagnosis

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Accurate differential diagnosis in gastroenterology requires the integration of longitudinal clinical narratives, similar cases, current biomedical literature, and specialty guidelines. Here we present **Medical Skill Hub**, an agentic clinical decision-support system that produces ranked diagnostic candidates with ICD-10-CM codes and traceable supporting evidence. The system combines an initial language-model assessment with hybrid similar-case retrieval, PubMed search, and locally compiled guideline skills. We evaluate the system across diverse digestive diseases using stage-specific and final Top-k recall. This template paragraph should be replaced with a concise summary of the clinical motivation, methods, principal numerical findings, and implications of the work.

1 Main

Digestive diseases present substantial diagnostic challenges because symptoms, laboratory abnormalities, imaging findings, and endoscopic observations often overlap across conditions. Clinical decision-support systems must therefore combine broad medical knowledge with patient-specific evidence while preserving traceability and clearly separating observed facts from retrieved information.

Recent progress in large language models and retrieval-augmented generation has enabled systems that synthesize heterogeneous evidence sources [1, 2]. However, diagnostic performance alone is insufficient for medical use: each candidate should be supported by verifiable evidence, and external knowledge must not be presented as an observed patient fact.

Here, we introduce **Medical Skill Hub**, a multi-stage framework for digestive disease diagnosis. Given a complete case narrative, the system generates initial hypotheses, retrieves similar cases, plans literature searches, consults local guideline skills, and produces five unique ranked diagnoses with ICD-10-CM codes. A diagnostic judgement stage determines whether a second evidence- gathering round is needed.

Our main contributions are:

- an evidence-grounded, multi-agent workflow covering the full spectrum of digestive diseases rather than a single condition;
- hybrid case retrieval that combines lexical and dense representations;
- traceable use of PubMed literature and locally compiled clinical guidelines; and
- stage-wise evaluation of diagnostic recall and evidence usage.

2 Results

2.1 System Overview

Figure 1 summarizes the diagnostic workflow. The preprocessing agent extracts positive clinical features and proposes initial hypotheses. A retrieval module ranks local cases using BM25 and dense similarity, fuses the rankings, and applies a medical cross-encoder. Search-planning, literature, and guideline agents then collect evidence for the final diagnostic agent.

Figure 1: Overview of the Medical Skill Hub workflow. Replace the placeholder with a vector PDF or high-resolution image. Describe each panel in order and define all abbreviations.

2.2 Evaluation Settings

Describe the study population, inclusion criteria, case sources, comparison methods, model settings, and primary endpoints. Keep training, validation, and test cohorts clearly separated. For clinical datasets such as MIMIC-IV [3], report governance and data-access requirements.

The primary metrics are category- and subcategory-level Recall@1, Recall@3, and Recall@5. Confidence intervals and statistical tests should be reported where appropriate. Complete baseline and system results are reported in Extended Data Table 2.

2.3 Overall Diagnostic Performance

Report the principal comparison first, including cohort size, effect magnitude, and uncertainty. Follow with results across disease categories, care settings, or case-complexity strata. Avoid restating every value shown in a table.

2.4 Contribution of Evidence Sources

Quantify the contribution of similar cases, biomedical literature, and local guideline skills. Distinguish evidence availability from evidence correctness, and document cases in which an external source could not be retrieved.

2.5 Ablation and Error Analysis

Evaluate key pipeline components and summarize clinically meaningful failure modes. Suggested groups include insufficient case information, competing diagnoses with overlapping presentations, coding granularity errors, retrieval failures, and unsupported model inferences.

3 Discussion

In this study, we present **Medical Skill Hub**, an agentic clinical decision-support system for principal-diagnosis ranking across digestive diseases. Rather than relying on a single language model or a single knowledge source, the system combines full-text case interpretation, hybrid similar-case retrieval, targeted PubMed search, and disease-specific guideline skills within a candidate- constrained diagnostic workflow. This design addresses a central challenge in clinical artificial intelligence: diagnostic reasoning requires not only broad medical knowledge, but also faithful preservation of patient information, selection of evidence that is applicable to the specific clinical context, and a clear distinction between observed patient facts and externally retrieved knowledge.

Existing medical knowledge systems only partially address these requirements. Conventional retrieval-augmented generation typically follows a query–document-chunk–language-model pipeline [2]. Although this approach can provide relevant background information, similarity-based chunk retrieval does not ensure clinical applicability. Guideline recommendations often depend on combinations of disease subtype, anatomical site, severity, disease stage, complications, previous treatment, and patient characteristics. These conditions may be separated across multiple chunks or omitted from the retrieved context, allowing an individually relevant passage to be applied outside its intended scope. Chunk-level retrieval can also obscure the relationship between a recommendation and its surrounding definitions, exceptions, and evidence statements. Knowledge graphs provide a more explicit representation of medical entities and relations, but their construction requires substantial schema design and continuing expert curation. Fixed graph relations may incompletely represent qualifiers and narrative context in clinical guidelines, while incorporating revised recommendations requires coordinated updates to nodes, relations, and downstream retrieval logic.

Phenotype-centered diagnostic systems introduce a different information bottleneck. DeepRare and traditional phenotype-matching tools represent patient findings using Human Phenotype Ontology terms, which facilitates standardized matching across diseases. However, converting a complete clinical narrative into a set of ontology terms is inherently lossy. Temporal progression, quantitative measurements, anatomical details, disease severity, procedural context, and interactions among findings may be simplified or discarded. Automatically extracted HPO profiles may also vary with extraction prompts, models, or terminology mapping. Because these profiles become the principal input to downstream retrieval and diagnosis, omitted or inconsistently encoded findings can propagate through the pipeline and produce unstable candidate rankings. Standardized phenotypes therefore remain useful retrieval signals, but they should not replace the original clinical narrative as the source of patient-specific diagnostic evidence.

Medical Skill Hub addresses these limitations by separating retrieval representations from diagnostic evidence. Structured positive features are used to improve similar-case and guideline retrieval, while the complete original case remains available to the diagnostic agents and is the only permitted source of facts about the current patient. This prevents the structured representation from becoming a lossy substitute for the record. Candidate diagnoses are generated independently by the language model and the similar-case module, merged by ICD-10-CM code, and subsequently constrained throughout evidence retrieval and final ranking. PubMed sections and previous cases are explicitly labelled as external evidence, reducing the risk that retrieved information will be incorrectly restated as an observed patient finding. The bounded second-round judgement further focuses retrieval on unresolved distinctions without allowing an open-ended reasoning loop.

The Medical Skill architecture extends this separation from patient representation to clinical knowledge representation. Instead of treating a guideline as an undifferentiated collection of retrievable chunks, each guideline is compiled into a disease-scoped module containing explicit scope metadata, operational instructions, a recommendation index, the complete source text, and a local retrieval procedure. At run time, skills are selected only when their disease scope and required qualifiers match the diagnostic candidates. The recommendation index supports efficient localization, whereas the full guideline text remains the final source used to verify context. The resulting workflow can be summarized as *guideline* → *structured clinical knowledge* → *disease-specific skill* → *conditional retrieval* → *evidence-grounded reasoning*. This organization preserves the traceability of the original document while adding clinical selection rules that are difficult to express through vector similarity alone.

Medical Skills also provide practical advantages for maintaining a clinical knowledge system. Because each skill is an independent module, a revised guideline can be recompiled and reviewed without rebuilding an entire knowledge graph or changing unrelated disease modules. Its metadata defines when the knowledge is applicable, its source text supports auditing, and its execution instructions define how an agent should retrieve and verify evidence. New disease areas can therefore be added incrementally, and different agents can invoke the same guideline knowledge without embedding the complete guideline library in every prompt. This modularity makes the system better aligned with clinical workflows, in which clinicians consult different specialty guidelines according to the current differential diagnosis rather than search all medical documents uniformly.

The findings suggest that diagnostic support benefits from combining complementary forms of evidence rather than treating language-model knowledge, retrieved documents, or historical cases as individually sufficient. Similar cases provide empirical precedents, PubMed retrieval addresses targeted knowledge gaps, and guideline skills supply clinically qualified recommendations. Their value lies not only in potentially improving diagnostic ranking, but also in making the basis for a ranking easier to inspect. The candidate constraint and explicit evidence provenance further reduce the space in which unsupported diagnoses or citations can be introduced, although they cannot eliminate errors caused by incomplete records, missing candidates, inadequate evidence, or incorrect clinical weighting.

In clinical practice, such a system could reduce the time clinicians spend searching guidelines, literature, and comparable cases, while preserving access to the underlying evidence for independent review. It may be particularly useful for non-specialist physicians who encounter unfamiliar digestive diseases or lack immediate access to subspecialty expertise. The system is intended to support, rather than automate, clinical judgement:

clinicians must still assess the completeness of the record, verify retrieved evidence, resolve conflicting findings, and determine the investigations required for confirmation. Prospective studies are needed to establish whether the proposed workflow improves diagnostic efficiency, consistency, and patient outcomes in real clinical settings.

4 Limitations

Several limitations define the scope of the present study. First, the current collection of guideline skills covers only a subset of digestive diseases and clinical contexts. A diagnosis for which no directly applicable skill is available must rely on similar cases, PubMed evidence, and the language model's internal knowledge. Even when a relevant guideline is present, its scope may not cover every disease stage, complication, procedure, or patient subgroup. The quality of a compiled skill also depends on the completeness of the source document and the accuracy with which its scope, recommendation index, and full text are represented. Future work should expand the skill library systematically, introduce expert review of compiled skills, and establish a versioned update process so that revised or superseded guidelines can be identified and replaced without compromising provenance.

Second, evaluation was limited by the availability of datasets containing complete case information suitable for diagnosis. Many accessible clinical datasets provide only structured variables, short case descriptions, isolated notes, or previously assigned diagnostic labels. Such data do not reproduce the longitudinal narratives, examination findings, laboratory trends, imaging, endoscopy, pathology, and clinical context that physicians integrate during diagnosis. Conversely, datasets containing sufficiently complete records are difficult to share because of privacy, governance, and institutional access constraints. The present results may therefore not represent the full diversity of digestive diseases, institutions, documentation styles, or patient populations. In addition, the recorded principal ICD-10-CM code is an administrative reference rather than an independently adjudicated diagnostic gold standard and may contain coding errors or omit clinically plausible alternatives. Evaluation on larger multi-institutional cohorts with clinician- adjudicated reference diagnoses is needed.

Third, the current system accepts textual case narratives only. Imaging, endoscopic photographs, histopathology slides, physiological waveforms, and raw genomic data must first be interpreted elsewhere and documented in text before they can inform the diagnostic pipeline. This restriction may discard visual or quantitative patterns that are difficult to describe completely and makes performance dependent on the quality of the written report. Extending the framework to multimodal models and modality-specific tools could allow the system to reason over source images and measurements while retaining the same provenance boundary between patient observations and external knowledge.

Fourth, the candidate-constrained design improves output consistency but places an upper bound on final diagnostic recall: a disease omitted by both the initial language-model hypotheses and similar-case retrieval cannot be introduced by later evidence retrieval. Similarly, relevant evidence may be missed because of imperfect search planning, incomplete PubMed indexing, or overly restrictive guideline-skill matching. The system also depends on external language models, PubMed availability, and the quality and representativeness of the local case bank. Changes to model versions or external databases may affect reproducibility and diagnostic ranking. Future studies should evaluate candidate-recovery mechanisms, alternative retrieval strategies, and performance under controlled changes to models and knowledge sources.

Finally, this retrospective technical evaluation did not measure how clinicians interact with the system or whether its evidence presentation reduces review time, changes diagnostic decisions, or improves patient outcomes. It also did not evaluate conversational collection of missing information from patients or clinicians. Prospective studies should assess usability, calibration, failure recognition, and the effect of the system on clinical workflow across different levels of specialist experience. **Medical Skill Hub** is intended only for research and technical demonstration. It does not replace clinician diagnosis, treatment decisions, or in-person medical assessment.

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List funding sources, institutional support, data providers, and individuals who contributed but do not meet authorship criteria. Include grant numbers where required.

6 Author Contributions

Use the CRediT taxonomy where possible. Example: F.A. and S.A. conceived the study; F.A. implemented the system; F.A. and T.A. performed experiments; all authors analyzed results, revised the manuscript, and approved the final version.

7 Competing Interests

The authors declare no competing interests. Replace this statement if any financial or non-financial interests must be disclosed.

8 Methods

8.1 Problem Formulation

We formulated digestive-disease diagnosis as principal-diagnosis ranking from a complete free-text clinical record. For a case x, the system returns an ordered set $\widehat{Y}_5 = \{(d_j, c_j, q_j, e_j, a_j)\}_{j=1}^{5}$, where d_j is a unique ICD-10-CM code, c_j is its canonical English name, q_j is an integer confidence score from 0 to 100, e_j contains case-grounded supporting evidence, and a_j contains recommended next examinations or management directions. The first-ranked item is intended to represent the condition chiefly responsible for the admission or the main condition evaluated and treated during the encounter. Codes are normalized to uppercase and stored without decimal points. Three- character categories are used only when a more specific supported subcategory is not available.

We enforced a strict provenance boundary throughout the pipeline. The original case text was the only permissible source of facts about the current patient. Initial hypotheses, retrieved cases, PubMed abstracts, guideline statements, and outputs from previous rounds were treated as candidate-generation signals or external knowledge, not as observations in the current patient. Consequently, the system could use external evidence to interpret a documented finding but could not import a symptom, test result, or diagnosis from another source into the patient's record.

8.2 Overall Diagnostic Workflow

The pipeline comprised six linked stages. First, two independent preprocessing agents generated an initial differential diagnosis and extracted structured positive findings from the original record. Second, the structured findings were used to retrieve diagnostically similar cases from a local case bank. Third, the five initial hypotheses and the leading similar-case diagnoses were merged and ranked to form a candidate set, and a search-planning agent generated PubMed queries for that set. Fourth, PubMed retrieval and local guideline-skill search were executed concurrently. Fifth, a diagnosis agent ranked the supplied candidates using the patient record and the retrieved external evidence. Finally, an evidence-sufficiency agent determined whether a second, targeted literature retrieval round was warranted.

The initial preprocessing and similar-case retrieval were performed once. The system allowed at most two diagnostic rounds. When the first-round evidence was sufficient, the pipeline returned immediately. Otherwise, the second round retained the original candidate set, similar-case results, and guideline results, but regenerated focused PubMed queries and accumulated newly selected abstract sections before revising the final ranking. This bounded design prevented an open-ended agent loop while preserving one opportunity to resolve a specific evidence gap.

8.3 Independent Clinical Preprocessing

Two agents processed the same original case in parallel and were not shown each other's outputs. The hypothesis agent independently produced exactly five unique principal-diagnosis candidates, each represented by a complete ICD-10-CM code and canonical English name. Chronic comorbidities, incidental findings, symptoms, and secondary complications were not used merely to fill lower ranks.

The feature-extraction agent organized explicitly documented positive findings into five fields: present illness history, past medical history, physical examination, family history, and pertinent laboratory, imaging, endoscopic, pathological, and microbiological results. Each finding was represented as a short English phrase while retaining clinically relevant duration, severity, anatomical location, measurements, and trends. Denied symptoms, normal or negative examinations, planned tests, and inferred findings were excluded. Empty fields were preserved as empty lists. These field-specific representations were used only as queries for similar-case and guideline retrieval; all subsequent diagnostic claims remained anchored to the full original record.

8.4 Similar-Case Retrieval

The local case bank stored one admission identifier, discharge diagnosis, ICD-10-CM code, and the same five clinical fields for each case. Non-empty fields were segmented into non-overlapping chunks of at most 510 tokenizer tokens. For the current patient, all items within each non-empty feature field were concatenated to create one field-specific query. Retrieval was conducted independently within the corresponding field of the case bank, avoiding, for example, direct comparison of a laboratory query with a family-history section.

We used parallel lexical and dense retrieval branches. The lexical branch applied BM25 [4] after lower-casing and tokenizing text with the scientific spaCy tokenizer, and retained up to 50 positive-scoring sections per field. The dense branch encoded each query and case section with BGE-M3 [5]. We used the normalized embedding of the first output token, truncated model inputs to 1,024 tokens, and ranked sections by the dot product of L2-normalized vectors, which is equivalent to cosine similarity. Up to 50 dense candidates were retained per field. Corpus embeddings and BM25 indices were cached and invalidated when the source case-bank fingerprint changed.

Section scores were aggregated at the admission level. For each field and retrieval branch, the case score was the best section score plus 0.2 times the second-best score when a second matching section existed. The five feature fields were assigned weights of 1.0, 0.5, 0.5, 0.2, and 1.0 for present illness, past medical history, physical examination, family history, and pertinent results, respectively. We combined ranks using reciprocal rank fusion (RRF) [6]. Let $\mathcal{F}$ denote the non-empty patient fields, $\bar{w}_f = w_f / \sum_{g \in \mathcal{F}} w_g$ the normalized field weight, and $r_{m,f}(i)$ the rank of case i for modality m and field f. The modality- specific fusion score was

$$S_m(i) = \sum_{f \in \mathcal{F}} \frac{\bar{w}_f}{60 + r_{m,f}(i)}, \tag{1}$$

where a term was omitted when the case did not occur in that ranked list. The hybrid score combined the lexical and dense branches as

$$S_{\text{hybrid}}(i) = \sum_{f \in \mathcal{F}} \bar{w}_f \left[\frac{0.4}{60 + r_{\text{BM25},f}(i)} + \frac{0.6}{60 + r_{\text{dense},f}(i)} \right]. \tag{2}$$

The fused ranking was deduplicated by ICD-10-CM code and truncated to 20 disease candidates. When more than five candidates were available, an LLM reranker was given anonymous candidate identifiers, diagnosis codes and names, BM25, dense, and RRF ranks, and the two highest-scoring fused sections. It selected exactly five unique candidates according to their fit with the current patient's structured findings. Similar-case text remained explicitly marked as external case evidence. If reranking failed validation or timed out after 120 seconds, the top five RRF candidates were used; when five or fewer fused candidates were available, their RRF order was retained directly. The pipeline also preserved the distinct top-five BM25, dense, RRF, and LLM-reranked outputs for analysis.

8.5 Candidate Integration and Search Planning

Candidate construction began with the five independently generated hypotheses, followed by up to five LLM-reranked similar-case diagnoses, with the top RRF cases used as fallback. Exact normalized ICD-10-CM duplicates were removed while preserving the first occurrence, yielding at most ten candidates. When at least two candidates were present, a separate planning reranker ordered all supplied codes against the original case. Exact, four-character, and three-character agreement between the initial-LLM and similar-case sources served as secondary signals and could not override contradictory patient findings. The reranker was required to return an exact permutation of the input codes; otherwise, the initial merge order was retained.

The search-planning agent copied this ranked candidate set without adding, removing, or renaming a diagnosis. In the initial round, it generated five to ten concise PubMed-oriented keyword queries whose collective coverage included every candidate and the most discriminative documented patient features. Queries could target diagnostic criteria, anatomical distributions, imaging, endoscopy, histopathology, procedural complications, or pairwise differential diagnosis. If a second round was requested, the agent instead generated one to five focused queries restricted to the diagnoses and knowledge gaps specified by the evidence-sufficiency judgement.

8.6 PubMed Retrieval and Evidence Selection

Each planned query was submitted to the PubMed E-utilities ESearch endpoint, and up to three identifiers were retrieved. The corresponding records were then downloaded from EFetch in XML format. Requests included the configured tool name, email address, and API key when available. A process-wide rate limiter enforced the configured request frequency; HTTP 429 and server errors were retried with exponential delay and jitter. Queries from the same planning result were executed concurrently.

The parser retained only records with a numeric PMID, non-empty title, and at least one abstract section. Structured abstracts were preserved as separate, one-indexed sections rather than collapsed into a single text block. A knowledge- search agent received all retrieved titles and sections and returned only exact (PMID, section index) pairs judged relevant to one or more planned queries. Repeated retrieval of a PMID across queries was treated as a relevance signal but was not sufficient for selection. The agent could neither rewrite an abstract nor invent a PMID or section index.

8.7 Local Guideline-Skill Retrieval

Clinical guidelines were prepared offline as local Agent Skills. Each skill contained disease-scope metadata, workflow instructions, a recommendation index, the converted guideline full text, and a local search utility. At run time, the guideline orchestrator compared the candidate diagnoses with the skill catalog. A skill was selected directly when its primary disease matched a candidate or when its broader scope clearly covered that candidate. A narrower skill was eligible only if all of its required qualifiers, such as subtype, complication, stage, hereditary status, or procedure context, were present in the candidate. Shared organ systems, symptoms, or treatments alone were not sufficient.

For every directly matched skill, the system performed one non-recursive forward expansion through differential diseases explicitly declared in that skill's metadata. An additional skill was selected only when its primary scope directly matched one of these declared differentials; reverse, multi-hop, and generic category expansion were prohibited. Selected skills were executed concurrently, with at most ten active executions. Each executor ran in a read-only sandbox, loaded exactly one selected skill, read its complete instructions, used the recommendation index to locate relevant passages, and verified those passages against the guideline full text.

Each executor returned the disease assessed, verified evidence, a concise patient-to-guideline comparison, and one of four applicability labels: *supported*, *possible*, *insufficient evidence*, or *not supported*. Empty guideline evidence necessarily produced the insufficient-evidence label. Before final diagnosis, directly matched skills were retained when labelled supported or possible, whereas differentially expanded skills were retained only when labelled supported. This stricter rule prevented a one-hop expansion from entering the final evidence set solely because it was compatible with the patient.

8.8 Candidate-Constrained Final Diagnosis

The final diagnosis agent received the original case, the candidate diagnoses, selected guideline assessments, and selected PubMed sections. Each candidate was annotated with its planning rank, source membership, initial-LLM rank, similar- case rank, and exact, four-character, and three-character cross-source agreement. Guideline evidence was numbered first, followed by PubMed evidence. Candidate metadata and numbered material were treated as ranking priors and external knowledge, respectively, rather than patient facts.

The agent was constrained to return exactly five unique codes from the supplied candidate set and to preserve each corresponding canonical name. Rank 1 was optimized for principal-diagnosis precision, ranks 1–3 for the strongest alternatives, and ranks 1–5 for plausible diagnostic coverage. The first initial LLM hypothesis was retained at rank 1 unless explicit patient findings favored another supplied candidate and cross-source evidence also supported that change. Lower ranks could add three- or four-character ICD diversity only when clinical support was otherwise comparable; diversity alone could not displace a stronger candidate. Unsupported etiological, anatomical, complication, or severity specificity was prohibited.

For each diagnosis, the output included an independent integer confidence from 0 to 100, supporting findings copied or concisely derived from the current record, and recommended next steps. Every planning candidate omitted from the top five had to be returned once with a patient-grounded reason for exclusion. Programmatic validation rejected new codes, altered names, duplicate diagnoses, ranks other than 1–5, missing excluded candidates, and empty exclusion reasons. One correction attempt was allowed after a validation error. Finally, only external evidence actually cited in a diagnosis, exclusion reason, or next step was kept, and citation numbers were compactly reassigned in order of first appearance.

8.9 Evidence-Sufficiency Judgement and Iterative Refinement

After each final diagnosis, a separate judgement agent reviewed the patient record, search plan, retrieved PubMed and guideline results, and complete ranked output. It requested a second round only when a specific unresolved distinction could materially change the principal-diagnosis ranking or ICD-10-CM selection, the available external evidence did not address that distinction, and another literature search could realistically resolve it. Uncertainty requiring a new patient-specific test did not qualify as an external evidence gap.

When further retrieval was requested, the judgement output had to identify unique focus codes from the current top five, specific evidence gaps, and PubMed query directions. The search-planning agent used this feedback to revise the queries but retained the original candidate set. During round 2, only PubMed search was rerun; similar-case retrieval and guideline-skill execution were not repeated. Newly selected PubMed sections were merged with round-1 evidence and deduplicated by PMID and section index. The diagnosis agent then reassessed all candidates using the accumulated evidence and the previous judgement.

If the second judgement still requested more evidence, no third retrieval round was initiated. Instead, the system performed one corrective final-diagnosis pass using the accumulated evidence, explicitly preserving unresolved uncertainty, avoiding unsupported code specificity, and reducing confidence when appropriate. The last completed diagnosis was returned as the pipeline output.

8.10 Failure Handling and Implementation

The orchestration was implemented in Python 3.10 with the OpenAI Agents SDK, and all inter-agent outputs were validated with Pydantic schemas. The diagnosis model was configurable between the OpenAI Responses API, a DeepSeek OpenAI-compatible endpoint, and locally served Qwen models. Native structured output was used when supported; compatible chat-completion models were instructed to return JSON objects. Model temperature was set to zero where supported.

The external PubMed and guideline branches were run with independent exception handling. Failure of either branch produced an empty stage result and a failure reason while allowing diagnosis to continue with the remaining evidence. Similar-case retrieval failure was handled in the same way. If search planning failed, the merged candidate list was preserved and PubMed queries were left empty. These fallbacks affected evidence availability but did not relax the final five-code schema or the constraint that patient facts originate only from the case record.

8.11 Baselines

We compared **Medical Skill Hub** with recent general-purpose and medical LLMs, three scales of the Qwen3.5 family, and representative medical agentic systems. The baselines were organized as follows.

Latest LLMs.

- **DeepSeek-V4-Pro** is DeepSeek's proprietary flagship reasoning model and supports configurable reasoning effort through its API [7].
- **GPT-5.6 Sol** is OpenAI's proprietary flagship model for complex reasoning and professional tasks; we used the `gpt-5.6-sol` API model [8].
- **Claude Opus 5** is Anthropic's proprietary high-capability model for long-horizon reasoning and agentic workloads; we used the `claude-opus-5` API model [9].
- **Baichuan-M2-32B** is an open-weight 32-billion-parameter medical reasoning model trained with domain adaptation and multi-stage reinforcement learning supported by a large verifier system [10].
- **DiagAgent-14B** is an open-weight medical diagnostic model optimized through multi-turn reinforcement learning in a virtual clinical environment [11].

Qwen3.5 model family. We evaluated **Qwen3.5-9B**, **Qwen3.5-27B**, and **Qwen3.5-122B-A10B** to measure the effect of model scale under a common model family. The first two are dense models, whereas Qwen3.5-122B-A10B is a sparse mixture-of-experts model with 122 billion total parameters and approximately 10 billion activated parameters per token [12].

Other agentic systems.

- **MDAgents** adaptively selects solo or group collaboration based on estimated task complexity and coordinates expert assessment, discussion, and review for medical decision-making [13].
- **DeepRare** is a multi-agent rare-disease diagnosis system that combines a central host, specialized agent servers, more than 40 tools, and external medical knowledge to produce ranked diagnoses with traceable evidence [14].

For standalone LLM comparison, each model received the same complete case text and diagnostic instruction and was required to return five unique principal diagnosis candidates with ICD-10-CM codes. No similar-case, PubMed, or local guideline evidence was supplied to these standalone models. Agentic-system outputs were evaluated under the same Top-k ICD-prefix criteria described below.

8.12 Evaluation

Batch evaluation used the recorded principal ICD-10-CM code as the reference standard. We evaluated seven outputs: the initial LLM hypotheses; BM25, dense, RRF, and LLM-reranked similar cases; the merged search-planning candidates; and the final diagnoses. Before comparison, reference and predicted codes were upper-cased and stripped of decimal points. Disease-category agreement required the first three characters to match, whereas subcategory agreement required the first four characters to match.

For a dataset of N cases and an ICD prefix length $p \in \{3, 4\}$, Recall@k was computed as

$$\text{Recall@}\,k^{(p)} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\Big(y_i^{(p)} \in \widehat{Y}_{i,k}^{(p)}\Big)\,, \tag{3}$$

where $y_i^{(p)}$ is the length-p prefix of the reference code and $\widehat{Y}_{i,k}^{(p)}$ is the set of corresponding prefixes among the top k predictions. Recall@1, Recall@3, and Recall@5 were reported for all stages; Recall@20 was additionally available for the RRF candidate list, and Recall@10 for the search-planning list. Metrics were summarized for each diagnostic round and for the final returned round. Guideline-skill usage rate and per-skill usage counts were reported separately.

9 Data Availability

Describe which data can be shared, where they can be accessed, and any clinical privacy or licensing restrictions.

10 Code Availability

Provide the repository URL, release identifier, license, environment setup, and instructions needed to reproduce the reported experiments.

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11 Figure Legends

Figure 1. Overview of the Medical Skill Hub workflow. Provide a self-contained legend, explain each panel, and define every acronym, color, symbol, and statistical annotation.

A Supplementary Information

A.1 Diagnostic Results (Table Version)

Extended Data Table 2 | Digestive disease diagnosis results (complete-case input)

Model	ICD-10-CM category			ICD-10-CM subcategory		
	R@1	R@3	R@5	R@1	R@3	R@5
Latest LLMs						
DeepSeek-V4-Pro	**64.06**	74.61	76.83	**44.87**	55.50	57.34
GPT-5.6 Sol	–	–	–	–	–	–
Claude Opus 5	–	–	–	–	–	–
Baichuan-M2-32B	44.50	55.13	57.64	9.45	13.36	15.35
DiagAgent-14B	36.90	49.30	53.43	3.84	6.42	7.08
Qwen3.5 Model Family						
Qwen3.5-9B	43.25	52.92	55.20	13.73	17.71	18.75
Qwen3.5-27B	55.28	71.00	73.51	28.41	37.12	39.63
Qwen3.5-122B-A10B	59.63	73.58	75.94	31.37	41.62	43.25
Other Agentic Systems						
MDAgents	–	–	–	–	–	–
DeepRare	–	–	–	–	–	–
Medical Skill Hub System						
Medical Skill Hub (DeepSeek-V4-Pro)	62.88	**81.11**	**86.13**	44.13	**66.94**	**73.06**

Note: All values are percentages (%). Results were evaluated on 1,355 MIMIC-IV cases. ICD-10-CM category and subcategory agreement used the first three and four code characters, respectively. Standalone LLM results use the initial five diagnostic hypotheses; the Medical Skill Hub row uses the final five diagnoses. Cases without a valid model output were counted as misses. Bold blue-shaded cells denote the best observed result in each metric; "–" indicates that a result was not available.